



Development of urban forest policy-making toward governance in the Republic of Korea

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ABSTRACT

This research attempts to analyze the emergence and development of urban forest policies at the national and local levels in the Republic of Korea. The Policy Arrangement Approach (PAA) is applied as an analytical frame to analyze changes in the urban forest policies of the central and local governments. The PAA offers four dimensions that can be used to describe and analyze the policy process: actors, power, rules of the game and discourse. The research findings indicate that a discourse on sustainable development in which the social functions of forests are taken into account substantially contributed to the creation of urban forest policies. This discourse contributed to the activities of actors and their power relationships and to the introduction of new rules relating to Korean urban forest management. In addition to public actors, private actors have also participated in creating and managing urban forests. Various partnerships among actors were formed for urban forest management. Civil society exercised its power to design and manage urban forests through increased voluntary participation. The legislation relating to urban forests functioned as a framework for urban forest policies at the national and local levels. Agreements acted as new rules governing the relationships among the actors who were involved in urban forest management. In conclusion, the four dimensions of policy arrangements relating to urban forest policy, and the interconnections among these dimensions, elucidate the emergence and dynamic development of urban forest management in the direction of governance at the national and local level in the Republic of Korea. In particular, discourse about forests was a substantive dimension of policy arrangements, and it influenced changes in the identity of the participating actors and their power. The discourse contributed to the establishment and development of rules for urban forest management. Thus this research provides strong evidence that the PAA helps understand dynamic changes of urban forest policy-making toward governance.

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Introduction

The urban population in the Republic of Korea (ROK) increased from 4 million in 1950 to 38.7 million in 2005; and the percentage of the population in urban areas increased nearly fourfold from 21% in 1950 to 81% in 2005 (United Nations, 2010). The increase of the urban population elevated urban policies to their current position as an important policy domain of the central and local governments. The rural areas surrounding urban centers were consolidated into urban areas in 1995. The urban–rural consolidation policy led to a rapid increase in the number of cities and in the amount of area they covered. Under this policy the area of urban forests has also increased from 0.5 million ha in 1993 to 2.4 million ha in 1999.

The Korean laws defines that urban forest includes not only street trees, school forests and recreation forests within urban areas defined by the Establishment and Management of Forest Resource Act but also urban parks designated according to the Act on Urban Parks and Green Spaces (Koo et al., 2013). Urban forests of consolidated cities cover 2 million ha, which constituted approximately 83% of total forests of all urban areas in 1999 (Seok et al., 2001). The increase in size and number of urban forests has caused forest policy to be recognized as an important agenda item in the ROK since the late 1990s.

Urban forest policy reflects several changes in forest policies in the ROK. Korean policies in all sectors, including the forest sector, have traditionally been formed and implemented through the centralized policy-making structure of the central government. Since local self-governance was introduced in 1990s, however, the role of local governments has expanded and the level of private sector participation in the policy decision-making process has increased

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(Lee, 1996). The urban forest sector followed this trend. Local governments have been establishing and implementing policies on urban forests including private and public urban forests. Citizens and NGOs that were interested in urban forests took part in urban forest management at the local level. The contents of Korean forest policies expanded and became more varied through the process of globalization. Recently, Korean forest sector policies have started to address global environmental issues, such as climate change (Lee, 2009; Park et al., 2010). These trends influenced the emergence and development of urban forest policies in the ROK. Taking this context into account, this paper attempts to describe and interpret the changes of urban forest policies in the ROK by focusing on the four dimensions (actors, power, rules of the game and discourse) and on the interactions among the dimensions. This paper tries to identify which among the four dimensions of policy arrangement caused urban forest management to become more responsive to social changes. In this way, the article aims for a better understanding of the dynamics of urban forest policies.

The paper starts with a description of the theoretical background, focusing on the Policy Arrangement Approach (Arts and Leroy, 2006) as an analytical framework. The subsequent sections present the research method and results relating to the four dimensions of policy arrangement. Finally, the interactions among the four dimensions of urban forest policies are discussed, and lessons are drawn for future urban forest governance.

Theoretical framework

Policy Arrangement Approach

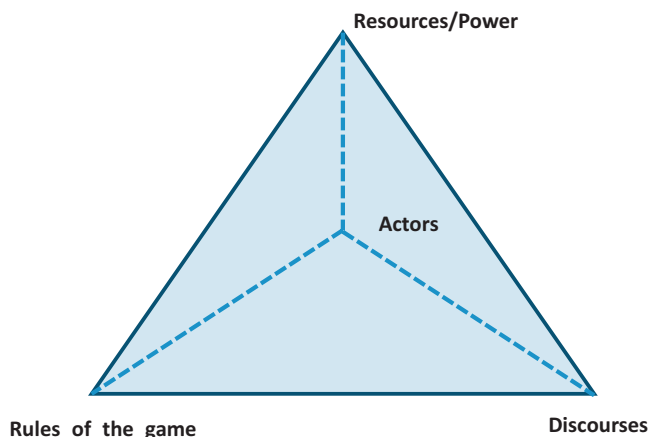
A policy arrangement has been defined as “the temporary stabilization of the content and organization of a particular policy domain” (Van Tatenhove et al., 2000, pp. 54). It includes structures formed through the processes as well as the interaction between policy actors and the formal and informal rules (Arts and Leroy, 2006). The Policy Arrangement Approach (PAA) is a method of policy analysis that was developed to help analysts understand changes of policy practices and structural changes in contemporary society (Liefferink, 2006, pp. 45). The PAA makes it possible to describe the dynamics of changes in the policy process within a given policy domain. The PAA offers four dimensions to describe and analyze the policy process: actors, resources or powers, rules of the game and discourse. The four dimensions form a tetrahedron, in which each corner represents one dimension, as shown in Fig. 1.

Actors consist of formal and informal actors who participate in the process of forest policy decision-making at various spatial levels. Policy actors play different roles in forming, implementing and evaluating forest policies in the policy cycle. In this research, the roles of the multiple stakeholders including central and local governments, NGOs, enterprises and citizens and the interactions and coalitions (Sabatier, 1998) among actors, are all described. The allocation of resources between the actors led to differences in power to influence the policy process. Resources to intervene or withdraw include money, information and political legitimacy. Rules of the game refer to formal procedures and informal routines of decision-making and implementation. Formal rules include the laws, ordinances and decrees that compose the legal framework. Voluntary agreements have come to play an important role as rules in forest management. Informal rules include customs and norms of the society. Discourse, as a substantive dimension, is “a shared way of apprehending the world” (Dryzek, 1997, pp. 8). It is defined as “an ensemble of ideas, concepts and categorizations that are produced, reproduced and transformed in a particular set of practices, and through which the meaning is given to physical and social realities” (Hajer, 1995, pp. 44). In relation to environment issues, discourse includes awareness and attitudes regarding environmental problems and approaches to solve the problems. Discourse on sustainable development and ecological modernization played a role in Europe as a major forest discourse from the 1980s to the 2000s (Arts et al., 2010).

In the PAA, the other three dimensions refer to the organizational aspects of policy, and the discourse dimension refers to substantial aspects of policy (Liefferink, 2006, pp. 47). The PAA has been applied to analyze various environmental policies, such as cultural heritage preservation in Norway, the United States and The Netherlands (de Boer, 2006); water management in the Netherlands (Wiering and Immink, 2006); marine policy in Belgium (Bogaert et al., 2009); health policy in the United Kingdom and Belgium (Stassen et al., 2010), and forest policy in the Netherlands (Veenman et al., 2009) and Belgium (Gossum et al., 2011). As these studies show, the PAA is an appropriate way to describe the dynamics of the change of policies with four interwoven dimensions.

Governance

The concept of governance emerged in political studies during the 1980s (Kjaer, 2004, pp. 1). Traditionally, governance was used as a synonym for government. Government is characterized by public actors' ability to make decisions, and its capacity to enforce them. In contemporary society, however, political decision-makers confront dynamic policy networks, risky issues, and new forms of political engagement (Hendriks, 2004). In this context, a new process of governing has emerged. Governance refers to “the development of governing styles in which boundaries between and within the public and private sectors have become blurred” (Stoker, 1998, pp. 17). The essence of this conceptualization is a focus on governing mechanisms that do not rest on the resources, authority and sanctions of government. Within a governance structure, self-organizing and inter-organizational networks are characterized by interdependence, resource exchange, rules of the game and significant autonomy from the state (Rhodes, 1997). There are many definitions of governance, but it generally involves various forms of partnership or networks among actors. The governance perspective pays attention to the increased involvement of the private sector and voluntary sectors in service delivery and strategic decision-making (Stoker, 1998). In the process, the responsibility of the government is shared, and the size of the government is reduced in public service. Governance networks create autonomous self-governing systems. Based on their authority, they use new tools and techniques to steer and guide public management. Especially in



Source: Liefferink (2006, p. 48)

Fig. 1. Tetrahedron representation of the Policy Arrangement Approach.

Table 1

Urban forests, urban forestry budget and population of major cities in the Republic of Korea.

	Busan	Daegu	Daejeon	Gwangju	Incheon	Seoul	Ulsan
Urban area (ha)	76,607	88,411	53,986	50,127	102,701	60,529	105,820
Urban forest area (ha)	15,922	24,173	32,096	24,278	12,631	13,714	8117
Urban forest area/urban area (%)	25.0	42.8	59.5	48.4	26.9	22.7	14.8
Constructed urban forest area (2005–2010) from KFS (ha)	45.08	67.39	82.33	92.90	7.60	109.10	14.62
Grants for construction urban forests from KFS (2005–2010) (million won)	7950	14,870	18,570	19,350	3278	13,047	3007
Population (1000 persons)	3393	2431	1490	1466	2632	9631	1071

Source: KFS (2010a), Statistic Korea (2010).

Note: 1 million Korean won was approximately equal to 880 USD in January 2010.

the environmental sector, voluntary agreements function as a new environmental policy instrument for environmental governance (Jordan et al., 2007). Voluntary environmental agreements among corporate, government and/or non-profit sectors aim to improve environmental quality, although they are not required by legislation (Karamanos, 2001). There are several forms of agreements at different levels, such as contracts and memoranda. In principle, there is neither a requirement to participate in an agreement nor penalties for non-participants. However, the obligation to draw up an agreement can be enshrined in law (Baggott, 1986, pp. 55). Therefore, in contrast to mandatory environmental regulations, voluntary agreements represent a type of institutional reform that can be called new public management (Kaboolian, 1998). In accordance with governance theory, this study investigates changes of forest policy administration within the Korean urban forest sector under the politics of urban forestry in the ROK.

Method

A case study method was used to investigate the dynamics of urban forest policies in the ROK. The Korea Forest Service (KFS) in the central government and in the local governments of seven metropolitan cities (namely the Metropolitan Cities of Seoul, Incheon, Daejeon, Daegu, Busan, Gwangju and Ulsan) were selected as the case study subjects. These cities are characterized by a different population size, urban area, urban forest area, and urban forest investments including constructed urban forest area (2005–2010) from KFS and grants for construction urban forests from KFS (2005–2010) (Table 1). Daejeon and Gwangju Metropolitan Cities with urban forests covering more than half of total urban area have established additional new urban forests with financial support by KFS than other cities during the period from 2005 to 2010. In contrast, Incheon and Ulsan Metropolitan Cities as industrial cities with relatively low rate of urban forest area have established a limited amount of new urban forests during the same period. In this research the similarities and differences of urban forest policies by the metropolitan cities will be described through the PAA approach.

For the analysis presented in this paper, documents such as plans, laws and ordinances, policy reports and research articles on urban forest management from 1990s to 2011 were collected and analyzed. Information written on forest governance was collected also from press releases and official websites of governments.

Information on urban forestry was categorized in relation to the four dimensions of the PAA. The ROK discourse on forest and urban forest was described on the basis of some key metaphors, which are frequently found in the relevant documents including forest related Acts, National Forest Plans and the Basic Plan for Urban Forest. The roles of public (ministries and cities) and private actors (NGOs, enterprises, scientists and citizens) in the urban forest sector and the relationships among them were also described and analyzed. In particular, the activities of new policy actors, such as forest associations and enterprises, were highlighted. Moreover, the powers of governmental agencies and of civil society in the urban forest sector

were described. Power was interpreted based on actors' resources including finance and expertise. In addition, the legal system as it relates as the rule for urban forests was investigated. Statutes at the national level and ordinances at the local level were described. Finally, agreements, which function as a new rule of the same in urban forestry, were also introduced.

Results

Discourse on urban forests

Forest policies of the ROK after the Korean War (1950–1953) can be divided into two phases. The first is oriented toward “forest protection”, focusing until the 1980s on reforestation of denuded forest lands. To restore the forests devastated by the Korean War, the ROK government established and implemented the first and second National Forestation Plans for the period from 1973 to 1987. The reforestation policy included planting trees and preventing illegal logging. It was linked to the Economic Development Plan, which had a top priority among the national policies promulgated by President Park Chung Hee. As a result of this policy, the ROK government completed reforestation of 2.6 million ha over a period of two decades (Bae et al., 2012, pp. 205). The ROK experience in reforestation has been recommended as an exemplary case to other developing countries (Noronha, 1981; Brown, 2008). The second phase of the ROK forest policies was oriented toward “sustainable forest management (SFM).” This orientation was explicitly reported as the basic principle for managing forestry in the text of fourth and fifth National Forest Plans (for the years 1998–2007, and 2008–2017, respectively) and in the Framework Act on Forestry enacted in 2001. The third National Forest Plan (1988–1997) embodied a transitional phase of the ROK forest policies, which started to pay attention to economic functions of forests. The fourth National Forest Plan introduced the concept of sustainable forest management, aiming at the enhancement of overall forest benefits including public welfare and recreational benefits. In accordance with the discourse about SFM in pursuit of maximizing the multiple functions of forests, urban forests were considered an ecological and social space for urban dwellers' recreation and health. With economic development, quality of life became more important as Korean society started to recognize the values of post-materialism (Ingelhart, 1997). City dwellers demanded more green areas for recreation and health within their urban environment. Focusing on social and recreational functions of forests, policies on urban forest management became relevant for enhancing the quality of urban life. Since the 1990s, the term “urban forest” has been mentioned in ROK policy documents. The Establishment and Management of Forest Resources Act enacted in 2005, for example, included articles on the establishment and management of urban forests.

The urban forest policies of the ROK reflect the global discourse on these issues. The discourse on sustainable development facilitated the discourse on sustainable forest management in the forest sector. The Forest Principles are a non-legally binding agreement on

forests signed by many countries at the United Nations Conference on Environment and Development (UNCED), which was held in Rio de Janeiro in 1992. These Principles recommend sustainable forest management to meet the social, economic, ecological, cultural and spiritual needs of present and future generations (United Nations, 1992b). Following this, KFS established and implemented policies toward sustainable forest management. UNCED also produced Agenda 21 as the means of giving real substance to sustainable development. Chapter 28 (Local Agenda 21) in Agenda 21 has precipitated extensive action of sustainable development at the level of municipalities (United Nations, 1992a). In ROK, 213 out of the 249 regional government units have adopted a Local Agenda 21 (Volkery et al., 2006). Korean Local Agenda 21 contributed to building environmental governance systems with multi-stakeholder process in political decision-making at the local level (Cha, 2008).

The urban forest policies of the ROK also reflect the global discourse on climate change. The fifth National Forest Plan (2008–2017) highlights the importance of forest functions in dealing with climate change based on global climate discourse. President Lee, Myung-bak, in his address to the nation on the occasion of the 60th anniversary of the founding of the ROK in 2008, embraced a vision of “Low Carbon, Green Growth (LCGG)” as a new paradigm to lead Korea’s future development for the next 60 years. After that address, LCGG was positioned as a guiding principle for the establishment and implementation of Korean policies in all sectors including the forest sector. Discourse on LCGG combines economic development discourse and environmental discourse. The Framework Act on LCGG defines the term of LCGG as follows: the term “low carbon” means lowering dependence on fossil fuels, expanding the use and distribution of clean energy, and reducing greenhouse gases to an appropriate or lower level with recognition of offsets by expanding carbon sinks (Article 2.1). The term “green growth” means growth achieved with efficient use of energy and resources mitigating climate change and reducing harm to the environment, securing new growth engines through research and development of green technology, creating new job opportunities, and achieving harmony between the economy and environment (Article 2.2). Under the national discourse on LCGG and the global discourse on climate change, national and local legal frameworks and plans regarding the expansion urban forests as carbon sinks have been established and implemented with the aim to mitigate and adapt to climate change in the ROK.

Actors: multiple stakeholders and their partnership

Urban forests are related to the forest sector and to the urban sector. Therefore, in the dimension of central governments as the center of the political system, the Korea Forest Service (KFS) and the Ministry of Land, Transport and Marine Affairs (MLTM) are in charge of urban forest policy-making in the ROK. The roles of the two organizations are differentiated. KFS develops and implements the basic plan for urban forests. It formulates the basic plans for urban forestry for the period of 2008–2017. The latter serve as the framework for national urban forest policies disclosed in 2007. The plan includes national strategies to manage and expand urban forests. The plan proposed the development of an information and management system for urban forests, networking urban forests, facilitating civil participation and improving the legal and financial system (KFS, 2007). It aims to facilitate urban forest policies of local governments in the ROK.

MLTM governs urban parks and green areas in cities. The 3rd National Land Development Plan (1992–2001) by MLTM as a Korean master plan of land management included national policy of green spaces such as increase of urban parks (Yang, 1996, pp. 91). In particular, MLTM has conducted major policies of urban parks and green areas as a part of urban development policy through

designing and managing Green Belts (which are restricted development zones) within urban areas as part of the national land planning and management system (Gallent and Kim, 2001; Bengston and Youn, 2006) in the process of Korean urbanization. Within local government, all metropolitan cities established Parks and Open Space Divisions of Environment and Forestry or Ecology Bureaus from 1991 to 1993 while Seoul pioneered in 1981 (Yang, 1996, pp. 92). These divisions have responsibility for urban policies including the expansion of green spaces within urban areas (e.g., planting and managing street trees, construction of green parks, and rooftop gardening).

Since the 1990s, with increasing attention of citizens on urban forests and urban green areas, various private actors have taken up conflicts over governmental plans of urban development. Urban citizens have conducted campaigns in several cities to protect urban forests that are threatened by government or corporate urban development plans (Park, 2009). Through these activities, coalitions of citizens and environmental NGOs, such as the Citizens’ Movement for Environmental Justice, have emerged. Environmental NGOs have supported the strategic activities of the urban forest movement that were conducted by citizens. Since the late 1990s, several forest NGOs were established, and they contributed to the establishment of urban forests. Forest for Life, a Korean forest NGO that was established in 1998, has led a movement for creating school forests by replacing the stone walls surrounding schools with forests. As a result, total 869 school forests were created through the participation of teachers, students and parents until 2010 (KFS, 2011, pp. 371). A private enterprise, Yuhan-Kimberly, which has conducted a long-term environmental campaign, has provided financial support for the school forest projects (Chung et al., 2011). The school forest movement thus shows how the governance structure relating to the urban forest policy can develop through the collaboration between NGOs, enterprises and citizens. Seoul Forest in Seoul Metropolitan City is another representative example of governance based on the increased involvement of the private sector (Stoker, 1998) in the ROK. In 2003, Forest for Life and Seoul Metropolitan City agreed on a Seoul Green Trust movement that was based on a partnership between Seoul Metropolitan City, enterprises and citizens. Citizens played an important role in the whole process of establishing new forests in Seoul, from the initial design stage to the construction stage, and finally to the on-going management of urban forests. They made a decision to establish the Seoul Forest Park in the site of a golf course and horse track at a location in the central part of Seoul City, and they followed through by donating money and planting trees. Therefore, the Seoul Forest Park was constructed and continues to be managed through public participation.

All metropolitan cities have committees on urban parks which consist of over ten representatives of city officials, experts such as scientists, citizens and NGOs, according to the ordinances on Urban Parks and Green Spaces. The committees as a network among various actors play a role of channel for communicating issues of urban forests and enable civil society to participate in designing and managing urban forests. These activities reveal multi-stakeholder process at the level of municipality under discourse on sustainable development including Local Agenda 21.

Rules of the game: acts, ordinances and agreements

In general, ROK forest policies are revealed in the process of the formulation and implementation of forest plans at the national and local levels. Several pieces of legislation give the central and local governments the responsibility to conduct forest planning and to implement forest plans (Fig. 2). KFS is required to establish and implement a national forest plan according to the Article 11 of Framework Act on Forestry (2001). The 4th National Forest

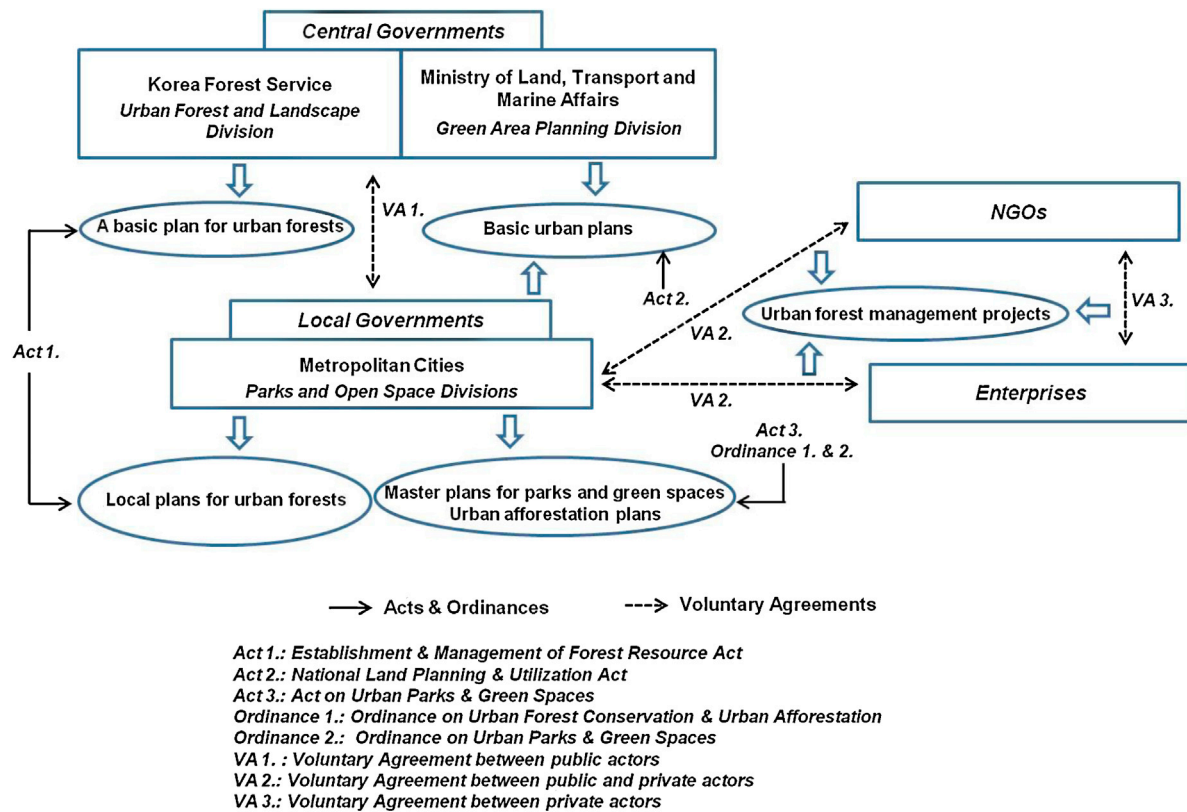


Fig. 2. Legislation and voluntary agreements as rules of urban forest management.

Plan for 1998–2007 and the 5th National Forest Plan for 2008–2017 include policies on establishment of urban forests for expansion of green space in urban areas. The 5th plan included the expansion of urban forests as a means of carbon sequestration as one of its 25 strategies.

In the ROK, urban forests are officially defined (by Article 2 of the Establishment and Management of Forest Resource Act of 2005) as “any forest and trees that are created and managed in a city for the health, recreation, emotional cultivation, hands-on activities, etc. of citizens”. In other words, urban forests encompass all wooded areas, including street trees and school forests, within the geographical area of cities. According to the Act, KFS and cities have responsibility for the establishment and implementation of plans for urban forestry. KFS should establish and implement the basic national urban forestry plan. In 2007, KFS did promulgate the first basic plan of urban forestry for the period from 2008 to 2017. The local governments in cities must also establish and implement local plans for the establishment and management of urban forests in the areas under their jurisdiction according to the national basic plan of urban forestry. Structurally, cities’ urban forest policies at the local level reflect the national policies formulated by KFS.

The “Act on Urban Parks and Green Spaces” established in March, 2005 requires the mayor of every metropolitan city to develop a basic plan showing a comprehensive direction for expanding, managing and exploiting parks and green spaces (Article 5). According to Article 6 of the Act, the master plan for parks and green spaces must be compatible with the basic urban plan based on National Land Planning and Utilization Act. Mayors should develop an urban afforestation plan for the part of the urban area under their jurisdiction in accordance with the master plan for parks and green spaces. Local ordinances on urban forests and parks support the implementation of the Acts. The Metropolitan Cities of Busan, Daegu, Incheon, Gwangju and Ulsan each established

an Ordinance on Urban Parks and Green Spaces, including urban afforestation plans, a real-name system of urban forest management, and tree-recording system. The Metropolitan Cities of Seoul, Incheon and Busan enacted Ordinances on Urban Forest Conservation and Urban Afforestation (Table 2). These ordinances support the contracts between the city and landowners for green conservation (Lee, 2002).

The discourse on LCGG influenced the legal system on urban forests. The ROK enacted the Framework Act on Low Carbon Green Growth as a Korean climate statute on January 13th, 2010 as the basic Korean legal framework on climate change policy. The Framework Act requires the government to establish a national strategy for LCGG (Article 9) and a basic plan every five years for coping with climate change over a planning period of 20 years (Article 40). In practice, the Task Force on Climate Change (which had previously been established under the Prime Minister’s Office) promulgated a Comprehensive Plan on Combating Climate Change in September 2008. KFS established the Comprehensive Forest Plan Combating Climate Change in December 2008 as an action plan for the forest sector in accordance with the Comprehensive Plan on Combating Climate Change. The forest plan recommends green policies including the expansion and maintenance of carbon sinks in forests. It includes alleviating the heat island effect through the establishment and management of urban forests and improving the soundness of the urban ecosystem (Lee, 2009). According to Article 11 of the Framework Act, each metropolitan city mayor must establish and implement a local action plan for green growth in conformity with the national strategy for green growth. To become a low carbon society, the Metropolitan Cities of Seoul, Incheon, Daejeon and Gwangju established Ordinances for Coping with Climate Change that have been in effect since 2008 (Table 2). All the climate ordinances recognize the significance of urban forests as carbon sinks, and they require policies for expansion of urban trees

Table 2

Rules of the games for urban forest policy of seven major cities in the Republic of Korea.

Rules	Busan	Daegu	Daejeon	Gwangju	Incheon	Seoul	Ulsan
Plans of urban forests (including urban parks and green spaces)	O ^a	O	O	O	O	O	O
Ordinance on urban forests (including urban parks and green spaces)	O	O	O	O	O	O	O
Ordinance on climate change	– ^b	–	O	O	O	O	–
Voluntary agreement with KFS	O	–	O	–	O	O	–
Voluntary agreement with private actors	–	–	–	O	–	O	–

^a 'O': established.^b '–': not established.

and forests. They include no detailed procedure for the expansion of urban forests as carbon sinks, but the articles relating to the expansion of carbon sinks through forests will play the role of a significant driver to establish and implement urban forest policies that address climate change.

“The district shall consider establishing and implementing programs to expand carbon sinks such as trees and forests to mitigate greenhouse gases.”

[Article 15, Ordinance Responding to Climate Change in Guro District, Seoul Metropolitan City]

Agreements or contracts play a role as new rules in the field of urban forest management. If it is for the purpose of expanding parks and green spaces, mayors may enter into contracts with the owners of any land in an urban area which has natural vegetation or a forest cover if the contract relates to support for the preservation, maintenance and utilization of such vegetation on the relevant land on the condition that the relevant land is made accessible to citizens (Act on Urban Parks and Green Spaces, Article 12). Mayors may also enter into contracts with the owners or the residents of the land in a certain urban area if the contracts call for the supply of saplings and support for necessary measures taken to supply them on the condition that they are used for the protection of forest green belts, an increase in the vegetation ratio compared to the area of the relevant land, and an increase in the vegetation that prevails over the relevant land (Act on Urban Parks and Green Spaces, Article 13). Several Metropolitan Cities have collaborated with private sector actors in forestation projects based on partnership agreements. In 2002, Gwangju Metropolitan City signed an agreement with a construction company, Namkwang for the construction of green streets in the city. The company voluntarily carried out the project for green streets using its own investment. In 2003, Seoul Metropolitan City and Forest for Life entered into an agreement to expand forest areas in Seoul and conducted the collaborative project for constructing Seoul forests. In 2011, Seoul Metropolitan City and the Chunhodong Catholic Church as a landowner made the first contract for green space use in ROK. The church provided residents with their private green spaces for public recreation areas through the contract. These agreements and contracts between the public sector and the private sector can be considered a type of new public management through institutional reforms (Kaboolian, 1998). Economic and budgetary restraints and demands for urban forests drove public management reforms (Wise, 2003) in urban forest management in ROK. Public management practices have changed from domination by bureaucracies to expanded participation by multiple actors. Agreements for public management have also been formed between government agencies at different levels (i.e., between agencies at the central level and agencies at the local level). Recently KFS and several metropolitan cities formed a coalition with the common goal of achieving LCGG in accordance with the climate discourse. The Metropolitan Cities of Seoul, Busan, Daejeon and Incheon signed a memorandum of understanding (MOU) with KFS that has been in effect since 2008 for low carbon, green growth city programs including expanding the carbon sink through forests. These agreements reflect vertical policy

integration (Lafferty and Hovden, 2003) within the urban forest sector to implement the overall policy objective of mitigating and adapting to climate change.

Power

The administrative power over urban forest policies in the ROK includes vertical and horizontal power relationships. In the horizontal dimension, the KFS as the relevant authority for the forest sector and the MLTM as the relevant authority for the urban sector have administrative power in relation to nationwide urban forest policies. As mentioned in the former section on actors, the KFS and MLTM have different approaches in relation to urban forests. In the administrative hierarchy, however, the KFS as a subordinate organization under the Ministry for Food, Agriculture, Forestry and Fisheries has less administrative power than the MLTM, which is a separate ministry. As municipal governments, metropolitan cities have responsibility for urban forest management including the designation and implementation of urban forest plans within the area of cities. All metropolitan cities established and implemented the master plans for urban parks and green spaces for next ten or twenty years according to the Act on Urban Parks and Green Spaces by MLTM between 2010 and 2011. The local plans for the establishment and management of urban forests according to the Establishment and Management of Forest Resource Act by KFS were included in the master plans for urban parks and green spaces as a part. This structure of planning indirectly reflects different power of the KFS and MLTM to municipal governments in the field of urban parks and forests. Functions of local governments have expanded since the local government reform and the formation of local councils in 1991. The enhanced administrative power of local governments, the increased ratio of local government employees to all government employees, and the increased share of local expenditures in the whole national budget (Lee, 1996), have all empowered the local self-governing system in various fields including urban forest management.

In the vertical dimension, the KFS as a central government agency influences the urban forest policies of cities. According to the Article 11 of the Framework Act on Forestry, the Minister of KFS may provide cities with different levels of budgetary support based on an evaluation of their performance in implementing the basic forest plans. Approximately 90% of the total budget for urban forests from KFS has been allocated to construction and management for urban forests in local governments (Table 3). In most cities, half of the budget for urban forest construction projects comes from the KFS and the other half comes from the local government revenue (KFS, 2010b). Therefore the grant by KFS actually influences construction activities of urban forests in the cities. Cities are also required to establish and implement plans for the creation and management of urban forests under the jurisdiction in accordance with the basic plan for urban forests promulgated by the KFS. Therefore, the KFS exercises power in relation to urban forest management in cities through their control of budgets and planning. This vertical structure contributes to maintaining consistency between the national and local urban forest policies. However, it

Table 3

Annual budget of Korea Forest Service for constructing and managing urban forests.

Budget (billion Korean Won)	2006	2007	2008	2009	2010	2011
Total	995.3	1204.0	1318.8	1547.4	1620.8	1674.2
Urban forest (%) ^a	22.1 (2%)	46.0 (4%)	67.2 (5%)	89.4 (6%)	89.2 (6%)	81.6 (5%)
National urban forests (%) ^b	4.2 (19%)	5.6 (12%)	4.7 (7%)	8.6 (10%)	9.7 (11%)	7.7 (9%)
Urban forests in local governments (%) ^b	17.9 (81%)	40.4 (88%)	62.5 (93%)	80.8 (90%)	79.5 (89%)	73.9 (91%)

^a The ratio of the budget of urban forest to total budget by KFS.^b The ratio of the budget to the total budget for urban forests by KFS.

hinders the development of more creative urban forest policies by cities in a way that could take the unique characteristics of cities into account. Cities lack financial and human resources for self-governing in the area of urban forest management. Occasionally urban development policies by local governments have not followed urban park and forest policies recommended by the central government, MLTM. Local governors and politicians preferred construction of roads, which can create jobs and boost land price than forest conservation such as design of green belts (Yoo, 2001). They emphasize economic development strategies to gain citizens' votes in the succeeding elections (Peterson, 1981).

As mentioned in the section on actors, the role of the private sector in the ROK urban forest management is increasing. Demands on urban forests by NGOs, citizens and enterprises have been a force for change in relation to land use policies in urban areas. Cases of conflicts between citizens and metropolitan municipality governments are worth observing. The residents of Mapo district in Seoul Metropolitan City stopped a plan by the city in 2003 to construct a water reservoir that would destroy Sungmi Mountain. In the case of Umyeon Mountain in Seoul Metropolitan City in 2003, citizens voluntarily initiated a national trust movement to collectively purchase forest areas to stop the threat of housing development projects (Park, 2009, pp. 133–134). In Daejeon Metropolitan City 293 associations and 698 individuals donated money and trees for city's campaign 'Planting 30 million trees in Daejeon Metropolitan City' for one year, 2007 (Daejeon Metropolitan City, 2008). In these cases, citizens played a role not only as observers but also as participants in urban forest management (Kim et al., 2006). In the case of the Seoul Forest Park in the Seoul Metropolitan City (Kim et al., 2009) and the Greenway Park in the Gwangju Metropolitan City (Jeon, 2009), in particular, citizens took part in designing and managing the urban forest parks. In the process of establishing the master plans for urban parks and green spaces in all seven metropolitan cities, the committees on urban parks and public hearings have comprised deliberative procedures to share information and opinions on urban forest management among multiple stakeholders. In particular enterprises and citizens offered financial support for constructing urban forests in Seoul, Daejeon, Ulsan and Incheon Metropolitan Cities (Kim et al., 2010). These phenomena show citizens' increasing power in policy decision-making in the urban forest sector. Civil society exerted their communicative power through campaigns, public hearings and advisory committees and administrative power through donation for urban forest maintenance and expansion, occasionally conflicting with urban development plans by municipalities and enterprises. The power of civil society has increased along with the decentralization process in the ROK. For three decades, South Korea had been ruled by military-dominated governments. The push for democratization and reform of politics and government increased and resulted in the creation of an anti-dictatorship coalition. This culminated in June 1987 in a pro-democracy protest against the military regime through the involvement of college students, opposition political parties, labor organizations, and lower- and middle-class citizens. This struggle against the authoritarian regime was named as "June Struggle". It can be characterized as a civic revolution from below,

in the sense that coordinated efforts of civic groups provided the major impetus for the country's democratic transformation (Oh, 1999). The year 1987 was "a turning point in which the locus of the critical driving energy shifted to the civil society from the state" (Lee, 1993, pp. 359). After the June Struggle in 1987, civil society was differentiated according to the interests of various groups. In the process, civil associations dedicated to the environment, education, health, human rights, women, and other concerns were organized. This change reflects the development of Korean democracy. While the Korean political system was undergoing democratization, the environmental movement was also gathering steam (Lim and Tang, 2002). After the democratic transition in 1987, many environmental organizations including forest NGOs were established in Korea. The Korean urban forest movement is based on this political background.

Discussion: interconnectedness among the four dimensions toward governance

In this research, the dynamics of urban forest policies were elucidated with particular attention to the interconnectedness of the four dimensions of policy arrangement in the ROK. The Korean discourse on urban forests formed in parallel to the change of discourse that was focused on economic development between 1970s and 1980s shifted to sustainable development in 1990s and LCGG in 2000s. Substantially, urban forest policies emerged from the discourse on SFM in which the environmental, social and cultural functions of forests are all taken into account. The forestry discourse shifted from forest rehabilitation to SFM, while increased concerns about quality of life with an emphasis on post-material values motivated the establishment and management of urban forests. The discourse on Local Agenda 21 regarding local sustainable development underscored the roles of local authorities and fostered participation by civil society in environmental decision-making in ROK. Recently LCGG, as a major theme in Korean climate discourse, offered a new perspective on urban forests as carbon sinks. The discourses, functioning as a substantive dimension, influenced the other three dimensions of PAA, namely actors, rules and power, of urban forest policy arrangement. The discourses on SFM, Local Agenda 21 and LCGG produced new institutional systems of urban forest management.

Central governmental agencies such as KFS and MLTM have guided local government to manage urban forests in accordance with the urban forest related plans that were mandated by the relevant legislation. After the 1991 local government reform in the ROK, metropolitan cities with enhanced administrative capacities have played an important role in the design and implementation of local plans for urban forests. The plans accommodate the relevant laws and ordinances that are based on the traditional legal framework. However, under the hierarchical administrative structure, metropolitan cities depend on the KFS's resources for establishing and implementing urban forest policies. Due to the limitations of their financial and human resources, it is difficult for cities to develop and implement creative policies of urban forests by themselves.

Table 4
Comparison between traditional and urban forest policies in the Republic of Korea.

	Traditional forest policies (Before 1990s)	Urban forest policies (Since 1990s)
Discourse	•Forest rehabilitation	•Sustainable forest management •Low carbon green growth
Actors	•Central governmental agency (Korea Forest Service) •Local governmental agency (cities)	•Central governmental agency (Korea Forest Service and Ministry of Land, Transport and Marine Affairs) •Local governmental agency (cities) •Civil society (citizens, NGOs, enterprises)
Rules	•Acts •Ordinances •Plans	•Acts •Ordinances •Plans •Voluntary agreements •Declarations
Power	•Hierarchical and centralized administrative power	•Hierarchical and decentralized administrative power •Local self-governing power •Communicative and administrative power through civil society

Private actors such as citizens, NGOs and enterprises have participated in the urban forest management process. Civil society, which has been empowered since the democratic transformation in 1987 brought about changes in urban land use plans. Sometimes this was done in conflicts and in collaboration with municipalities in other times civil society conducted various voluntary activities for the creation and management of urban forests, such as donating funds and monitoring performance. Its opinions and views on urban forest managements have been collected and reflected on the plans of urban parks and green spaces through committees on urban parks in all seven metropolitan cities. In Gwangju and Seoul Metropolitan Cities voluntary agreements between parties in the public and private sectors relating to urban forest management have functioned as new rules in urban forest management. This is illustrated by the introduction of New Public Management (Kaboolian, 1998) in the history of Korean forest policies. The contracts and agreements that function as new policy instruments have diversified the roles of political actors, and the power of civil society has increased in the political system relating to urban forest management policies. This phenomenon shows how urban forest policies reflect the institutional change of forest policies from centralized administration dominated by the bureaucracy to increased public participation by multiple actors. In conclusion, comparing with forest policies before 1990s, urban forest policies since 1990s include several changes such as participation by civil society, introduction of voluntary agreements and local self-governing system. This reflects transition of forest policies toward governance in ROK (Table 4). Therefore Korean urban forest policies generally proceed toward governance in the process of institutional change of forest policies. This research provides strong evidence that the PAA helps understand the institutional changes of urban forest management toward governance.

This research has some limitations in terms of the research structure and the interpretation of the research results. In the PAA, the dimensions of policy arrangements are functional in describing institutional changes. However, this analytic approach does not, in itself, explain why policies change, and it does not indicate the magnitude of these changes that have occurred (Wiering and Crabbé, 2006, pp. 94). It does not include causes of the differences of the forest policies among the different metropolitan cities. In spite of these challenges, however, the research findings do contribute to a better understanding of the development of urban forest policies and their dynamic changes in the ROK. Based on these findings, further studies on the forces of institutional change and their nature and magnitude in relation to urban forest policies as well as comparative research on urban forest policies at the national and local levels can be developed.

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